

DIABETIC RETINOPATHY SCREENING REPORT

AI-Assisted Fundus Image Analysis — with Bootstrap 95% CI

Image: idrid_IDRiD_0195.jpg
Date: 2026-03-28 18:04:23
Model: EfficientNet-B3 (Binary + Multiclass Cascade)
Bootstrap: 2000 resamples, 95% confidence level

SCREENING RESULT (with 95% Bootstrap CI)

Binary Detection:

- Ensemble (5-fold+TTA): DR Detected
P(DR) = 0.9863 95% CI [0.9841, 0.9886]
- 2-Phase (single+TTA): DR Detected
P(DR) = 0.9999 95% CI [0.9999, 1.0000]

Severity Grading (Direct Multiclass Model):

- Predicted Grade: 2 — Moderate NPDR
- Confidence: 69.1% 95% CI [0.5997, 0.7908]
- Severity: Moderate

Per-Grade Probabilities (95% CI):

G0 (No DR)	): 0.0134	[0.0091, 0.0183]
G1 (Mild NPDR)	): 0.0648	[0.0527, 0.0763]
G2 (Moderate NPDR)	): 0.6906	[0.5997, 0.7908] ◀
G3 (Severe NPDR)	): 0.1946	[0.1037, 0.2797]
G4 (Proliferative DR)	): 0.0365	[0.0190, 0.0603]

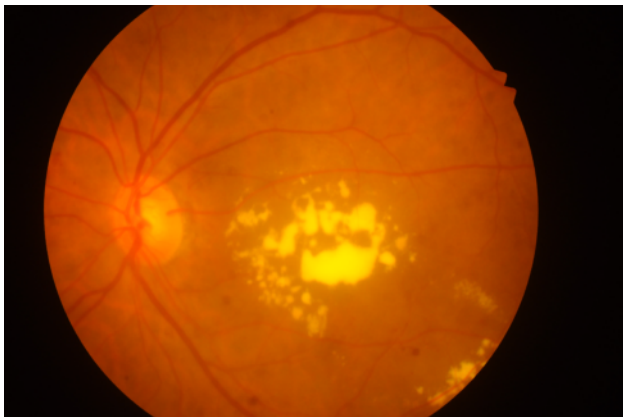
Clinical Recommendation:

- Refer to ophthalmologist within 3-6 months

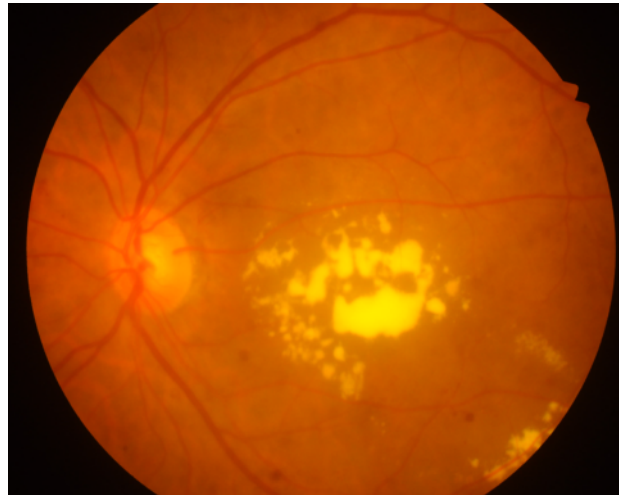
DISCLAIMER: This is an AI screening aid, not a diagnostic tool. All results must be confirmed by a qualified ophthalmologist.

Preprocessing Pipeline — idrid_IDRiD_0195.jpg

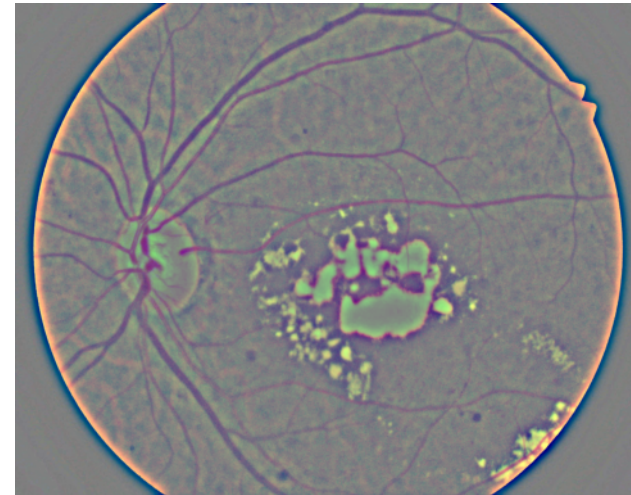
Step 1: Original



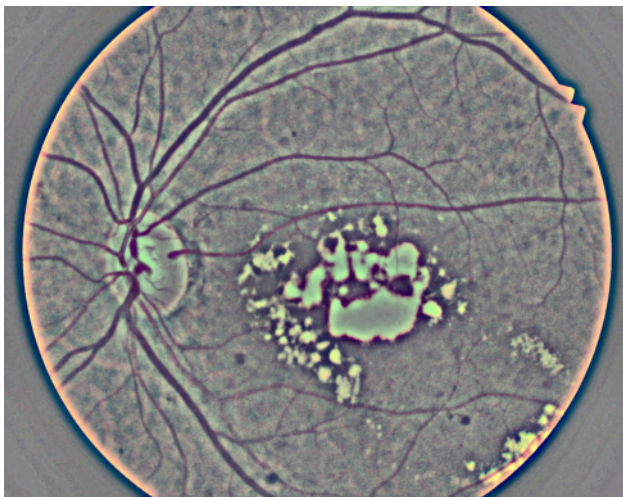
Step 2: Circle Crop



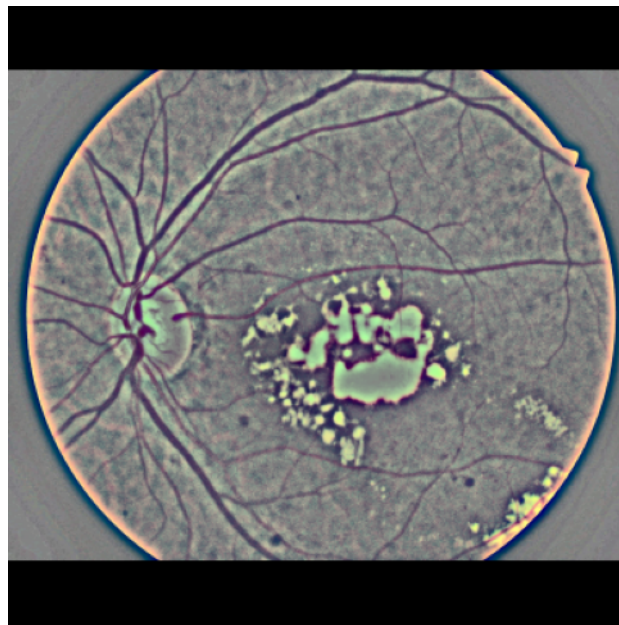
Step 3: Ben Graham



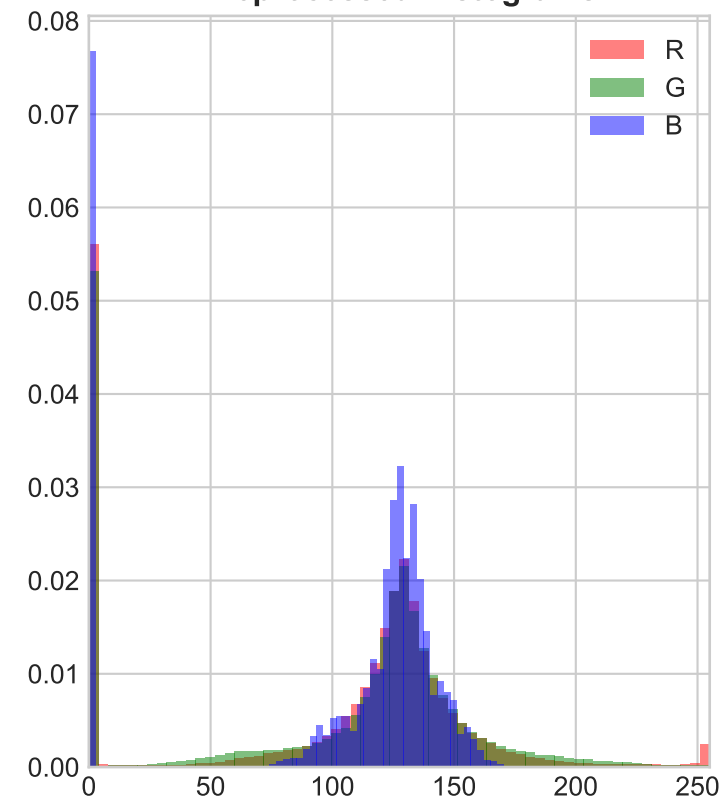
Step 4: CLAHE



Step 5: Final (384×384)

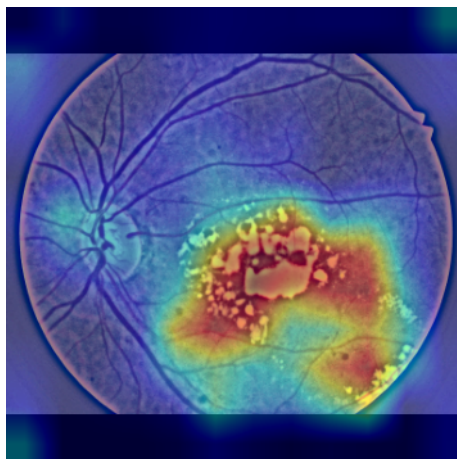


Preprocessed Histograms

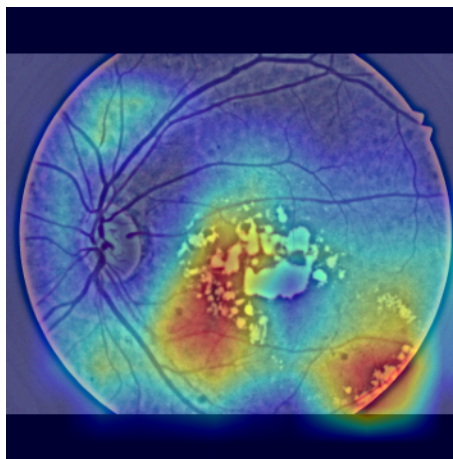


Explainability & Predictions with 95% CI — idrid_IDRiD_0195.jpg

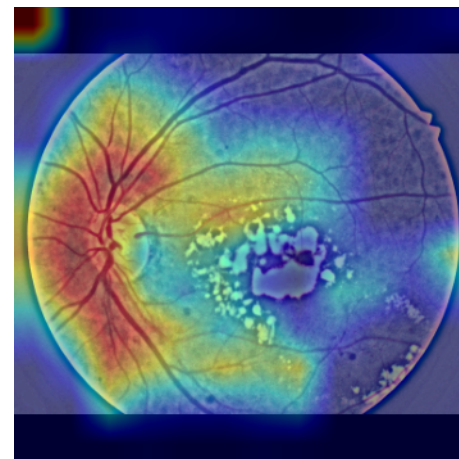
Ensemble GradCAM
P(DR)=0.9863 CI[0.9841,0.9886]



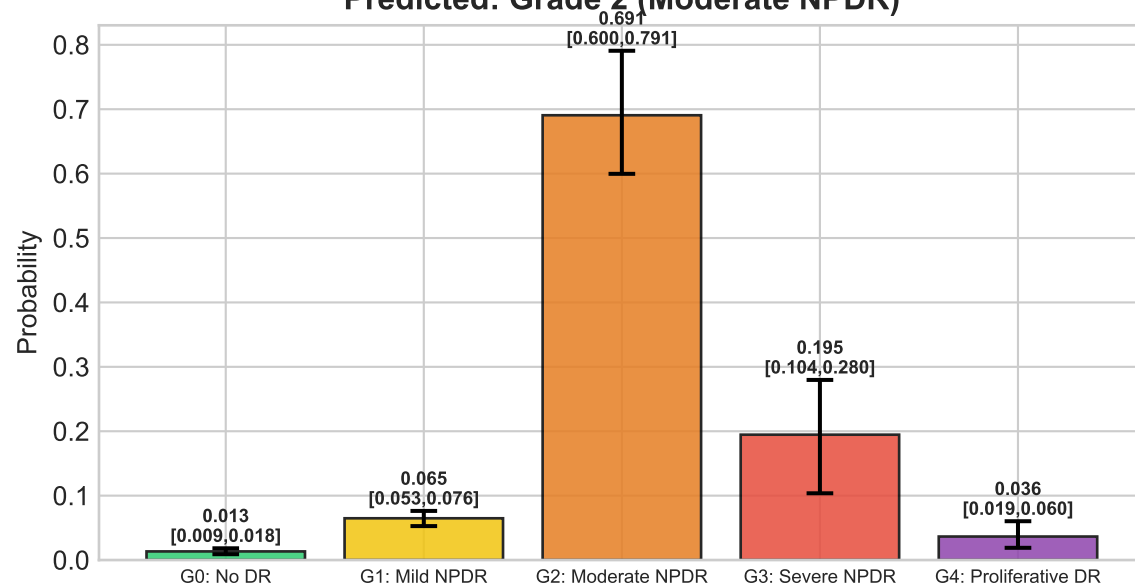
2-Phase GradCAM
P(DR)=0.9999 CI[0.9999,1.0000]



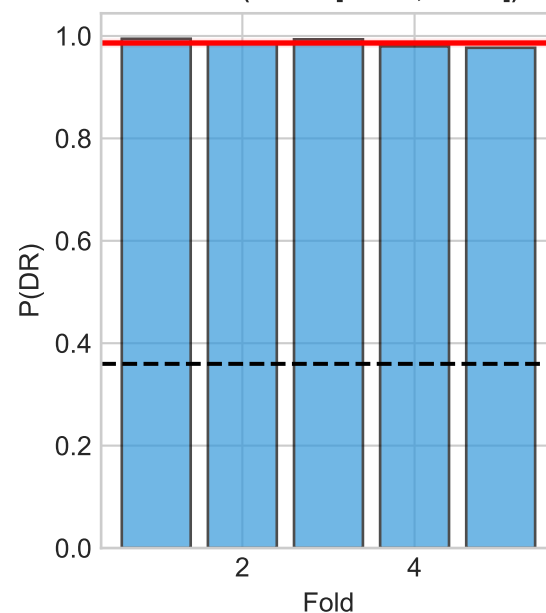
Multiclass GradCAM (G2)
Conf=69.06% CI[0.600,0.791]



5-Grade Probabilities with 95% CI (n_boot=2000)
Predicted: Grade 2 (Moderate NPDR)



Per-Fold (95% CI [0.9800, 0.9926])



Model Validation Summary

MODEL VALIDATION METRICS

Dataset: APTOS 2019 (531 test images)

Bootstrap: 2000 resamples, 95% CI

Metric	Ensemble+TTA	2-Phase+TTA
AUC-ROC	0.9992	0.9992
Accuracy	0.9906	0.9887
Sensitivity	0.9962	0.9962
Specificity	0.9852	0.9815
F1-Score	0.9905	0.9886
Cohen Kappa	0.9812	0.9774

Multiclass Model (Direct 5-Class):

Best Validation QWK: 0.859

Architecture: EfficientNet-B3 + Dropout(0.4) + Linear(1536, 5)

Training: CrossEntropyLoss + label_smoothing=0.1

THIS IMAGE – PREDICTION SUMMARY

Ensemble P(DR): 0.986295 95% CI [0.984067, 0.988617]

2-Phase P(DR): 0.999936 95% CI [0.999903, 0.999967]

Multiclass Grade: 2 (Moderate NPDR)

Confidence: 0.6906 95% CI [0.5997, 0.7908]

Per-Grade Probabilities (95% CI):

G0 No DR : 0.0134 [0.0091, 0.0183]

G1 Mild NPDR : 0.0648 [0.0527, 0.0763]

G2 Moderate NPDR : 0.6906 [0.5997, 0.7908]

G3 Severe NPDR : 0.1946 [0.1037, 0.2797]

G4 Proliferative DR : 0.0365 [0.0190, 0.0603]